Robots

Hello everyone, and welcome to this second episode of the Cottongue podcast.

[00:00:11] I'm very happy to be back with you for this second podcast. I hope you're well. I hope you're feeling good.

[00:00:20] In the previous podcast, we talked about languages and the theory of the American professor Stephen Krashen. I gave you some advice on being able to use a language. Do you remember my advice? My main piece of advice was to look for interesting things to read, watch or listen to in French. I advised you to forget about grammar and focus on comprehension. If every day you try to understand something in French — a text, a video or the lyrics of a song — you'll make enormous progress. And the most important thing for staying motivated is to find things that interest you, things you want to understand.

[00:01:21] Today, in fact, I suggest listening to a podcast in French about a subject I find fascinating. That subject is robots. So why do robots fascinate me? Because they're starting to live with us, to surround us. "Surround us" means that they're present around us: at work, when we go shopping, a little bit everywhere, really. But haven't you noticed them? Haven't you noticed the robots that are all around us? Perhaps that's because when I say the word "robot", you think of robots from science-fiction films. But a robot doesn't necessarily look like that. A robot can simply be a mechanical arm, like the arms we find in factories. You know, factories are places where products are made in very, very large quantities. For example, a car factory or a mobile-phone factory. In these factories, robots have replaced people, and the robots we find in these factories don't really look like the robots we see in science-fiction films.

[00:02:54] So what exactly is a robot? A robot is an automaton. It's a machine programmed to carry out tasks in place of people, to replace people. I told you about factories. Well, in factories, in car factories, for example, which produce cars, robots have almost completely replaced people. On the production line, we mainly find robots. That isn't really new. What's new, the innovation, is that robots can now interact with us. We can talk to them, to robots. They can provide services for us, analyse us and understand us.

[00:03:55] How are we going to live together with robots? How will we share our lives with them? Can robots replace all of us? Can we imagine a world in a few years' time, a future, in which there will be only robots? That makes us think of the plot of a science-fiction film, a disaster scenario. But today, we're going to try to provide somewhat more serious answers, perhaps, to see how we live with robots. We'll try to answer all these questions together.

[00:04:45] Are you ready? Then here we go!

[00:04:52] Imagine you're at home, in your own home. Your working day is over, and you're a little tired, a little bad-tempered. You've had a fairly unpleasant day. Perhaps your boss wasn't nice to you; perhaps you had a meeting that was really very, very boring. In short, your working day is over, you're at home, and you're in a rather bad mood. And then Pepper, your domestic robot, comes to see you. Pepper approaches you and asks whether you're all

right. So you answer Pepper: “Listen, Pepper, I’m not feeling very good today. I’ve had a bad day.” And then Pepper starts asking you questions to find out exactly what’s wrong. A little like a real human being. A little like a friend who was with you and asked you questions to try to cheer you up. “To cheer someone up” means “to try to give them their energy back so that they’re in a good mood.” For example, if you’re sad, you can watch a film, a comedy, to cheer yourself up, to give yourself some energy again and feel better. So you’re at home, and your robot Pepper tries to cheer you up. For example, it suggests watching a film or playing a game with it; it shows you funny videos on YouTube. And thanks to that, thanks to Pepper, after a few minutes you already feel better, you feel much better.

[00:07:15] Is this scenario difficult for you to believe? Is it difficult to imagine? Do you feel as though you’re in a science-fiction film?

[00:07:28] Well, nevertheless, Pepper really exists. Pepper is a robot produced by a French-Japanese company called Softbank Robotics. Pepper is present in some banks in Japan to help customers, to entertain customers, to ask them questions and to try to offer them solutions. Pepper robots are also present in some railway stations in France to help travellers who have questions too.

[00:08:15] These robots belong to a new category of robots, a new type of robot. They’re called “emotional robots”. So how do these emotional robots work? Emotional robots are able to recognise the main human emotions, to analyse and understand human emotions, and depending on this, they adapt their behaviour to interact with their user, with the person they’re communicating with, meaning the person who is talking to them.

[00:09:07] It may be a little difficult to imagine that a robot is capable of understanding your emotions, understanding your feelings. How do you think these robots identify your emotions? In fact, they have several sensors. Sensors are things that allow robots to record information. For example, they have a visual sensor that allows them to see, like humans. They have auditory sensors that allow them to listen, a little as if they had ears. They can listen to your answers and analyse that information. They can use their visual sensors, their eyes, to analyse your facial expressions, to see whether you’re smiling or laughing. And they can also analyse your voice. If you’re sad, it can be heard in your voice. You don’t have exactly the same voice, you don’t speak in exactly the same way when you’re sad and when you’re happy. Pepper can use all this information to understand your emotions, to identify your emotions.

[00:10:54] What’s more, Pepper wants to try to get to know you better, a little like a friend. That means it will ask you questions and remember, recall, your answers. For example, if you say to Pepper: “Pepper, I don’t like playing chess.” Well, Pepper will never suggest playing chess. Or if you say to Pepper: “Listen, Pepper, I don’t like watching films that are horror films,” then Pepper will never suggest watching horror films. That means Pepper learns to get to know you and develops with you, little by little. Pepper memorises your personality traits, your preferences, and adapts to your tastes and habits. So, ultimately, Pepper is a companion. It’s something that keeps you company.

[00:12:20] So if you want to buy a Pepper, you'll need to save up, because Pepper isn't cheap. When we say that something "isn't cheap", it means it's rather expensive, quite expensive. For example, we can say that a car isn't cheap, meaning that this car is expensive, that it costs a lot of money. How much does Pepper cost? Well, Pepper costs about €1,500. €1,500 plus a €200 subscription that you have to pay every month. This subscription is for servicing, maintaining and repairing the robot. There is someone who can come and repair your Pepper if your Pepper breaks down.

[00:13:27] Do you think it's possible for a robot to understand human emotions? We might think that human emotions, human feelings, are things that are very, very complex. Philosophers have been trying to understand human emotions for centuries and centuries. And yet we have the impression that some emotions are still a mystery. So if philosophers cannot understand our emotions, how could a robot be capable of doing so? For many people, this is very difficult to imagine. They think it's impossible. They think human emotions are too complex to be understood by a robot.

[00:14:27] But I think it is possible. I think it's completely possible. There is a limited number of emotions: joy, sadness, fear, and so on, and so on. These emotions aren't infinite. There aren't billions and billions of emotions. We could make a list of these emotions. And some emotions that are a little more complex are simply a mixture of basic emotions, fundamental emotions. If we think of nostalgia, nostalgia is when you think about a past period of your life and miss that period a little. For example, you're nostalgic for the years when you were a child, when you could do whatever you wanted, when you didn't have to do the cleaning, look after the home or work. Perhaps you're nostalgic for that period. Well, nostalgia is simply a mixture of regret and sadness. So this emotion isn't all that complex. And there isn't an unlimited number of emotions.

[00:16:04] What's more, we can assign a number of parameters to each emotion. When we cry, when we shed tears, generally it means that we're sad. But it can also mean that we're very, very happy and that we find something extremely funny. There is an expression in French: "to cry with laughter". Crying with laughter is when you watch a film that is so funny that it makes you cry: you cry with laughter.

[00:16:48] So we can associate parameters with each emotion, and then robots can analyse these different parameters to understand your emotions. In short, in my opinion, in the future robots will be able to understand us very, very well. They'll be able to understand people very well, and perhaps even understand people better than people understand themselves.

[00:17:29] Besides the Pepper robot, Softbank Robotics produces other types of robots with other functions. I'm going to tell you a little about these other robots. Another robot from Softbank Robotics is a robot called Roméo. Roméo is a humanoid robot, meaning that it resembles a human being, a person. It has a head, a body, two arms and two legs, and it measures 140 cm: it's 140 cm tall. Roméo's function, Roméo's purpose, is to help older people or people with disabilities. How can Roméo help older people or people with disabilities? It can help them open a door, climb a staircase or pick up objects on a table. Roméo can help these people in their daily lives to do things that are fairly difficult for them

because, for example, they cannot move around easily. But Roméo can move around easily, and it can go and get objects and bring objects to them.

[00:19:12] Do you think this is a good idea for helping older people or people with disabilities?

[00:19:20] On the one hand, it may be a good idea because in many countries there aren't enough people to help older people or people with disabilities. So this robot can be a real help to these people in their daily lives.

[00:19:44] But on the other hand, we might think that with these robots, we'll perhaps abandon older people and people with disabilities because we'll think: "OK, these people have robots to look after them, so they don't need people, human beings, to look after them." And perhaps that will increase their loneliness. Perhaps these people will feel alone.

[00:20:27] The third robot from Softbank Robotics is called Nao. Nao is smaller than the other robots from this company because it measures 58 cm. And Nao is used for education. Nao is able to move, sense, hear, speak, see and, of course, connect to the internet. It is used, for example, in schools where there are autistic pupils. Autism is a behavioural disorder that changes social interactions and communication. Generally, autistic people have difficulties expressing themselves and communicating, and they have difficulties with their everyday social interactions. Well, Nao encourages autistic children to express themselves. Nao is very patient and can listen to autistic children and encourage them to express themselves, to talk to it.

[00:21:58] Can you imagine robots that, in the future, will replace teachers, robots that will take the place of teachers?

[00:22:12] There would be several advantages to this situation. For example, robots are never ill, they're never tired, unlike teachers. The quality of lessons would always be the same. Robot teachers can work 24 hours a day, and they never make mistakes. What's more, they don't need to be paid. You only have to buy them, but after that they don't have a salary. So robot teachers would be cheaper than human teachers. But what do you think the disadvantages are? Perhaps these robots, these robot teachers, won't be able to answer every question. And they won't be able to build relationships with students either. And for students, it's important to have a connection with their teacher, to feel that their teacher knows them, that they can talk to their teacher freely, that they can communicate with their teacher.

[00:23:36] But we don't know. Perhaps in the future, there will be robot teachers in universities.

[00:23:49] To finish this podcast, I'm going to tell you about one last type of robot that, in my opinion, is extremely dangerous. These robots are autonomous weapons. What is an autonomous weapon? It's a robot used in an army, and it's like a soldier, a member of the military. These robots are used to wage war. For example, drones are used in Afghanistan and Iraq. These drones can already fly on their own. They don't need to be controlled by people. We can specify a location, tell them to go to a particular location, and the robots will

fly there on their own: they're autonomous. In a few years, they'll be able to kill human targets, to kill human beings, people, on their own, without intervention, without being controlled.

[00:25:02] That obviously raises ethical problems because in a few years, robots will be able to kill people, as in disaster science-fiction films. And robots don't experience feelings; they have no empathy. In other words, robots can kill soldiers or civilians without distinction. For a robot, it's the same. Because robots have no empathy, they don't share human emotions. If we tell a robot to kill someone, that robot will kill that person and won't reflect on its action. So that is a real moral problem, a real ethical problem.

[00:25:59] In fact, there are many associations of scientists, for example an association involving the very famous British physicist Stephen Hawking, that are calling for a ban on autonomous weapons. They want these robots to be banned. Other types of weapons are already banned, such as chemical weapons. Well, these associations of scientists want there to be no autonomous weapons. And I agree with these scientists. I think these autonomous weapons, these autonomous robots, represent a threat like nuclear weapons, for example: they're just as dangerous.

[00:26:53] To conclude this podcast, we can say that life with robots depends on how companies, the companies that produce these robots, will programme them. If companies produce robots to help us in our daily lives, that can be a very good idea. It can be a real help, for example for people with disabilities, for older people, in the same way that robots have been a great help in industrial production. But robots can also be very dangerous if we use them to wage war, for example. So that should be banned. It's the wrong direction; it's a bad use of robots.

[00:27:55] That's it: this is the end of this podcast. Thank you all for listening to it. I hope you found it interesting. I hope you had a good time. If you like, you can find the transcript of this podcast on my website, cottongue.com, with all the information and all the vocabulary. As usual, I encourage you to listen to this podcast several times to try to understand as much as possible, and I promise you that if you keep doing this, you'll make enormous progress, you'll understand more and more things, and you'll learn to express yourself naturally in French.

[00:28:52] In the next episode, we'll talk about happiness and the countries where people are happiest. Thank you, everyone. Thank you for listening to this podcast, and we'll meet again next week with a new episode. See you soon. Bye!